

# Chi-Square Tests for the Final Exam

MAT 1272 Statistics

# Learning Objectives

By the end of this review, you should be able to:

- Identify a Chi-Square Goodness-of-Fit Test.
- Identify a Chi-Square Test of Independence.
- State the null and alternative hypotheses.
- Compute the degrees of freedom.
- Recognize that chi-square tests are right-tailed.
- Use the TI-84 to obtain a p-value.
- Write a conclusion in context.

# Two Types of Chi-Square Tests

## 1. Goodness-of-Fit Test

- One categorical variable.
- Compare observed counts to expected percentages.

## 2. Test of Independence

- Two categorical variables.
- Data are presented in a contingency table.

# Goodness-of-Fit Test

Use this test when there is only one categorical variable.

Examples:

- Cereal brands
- Causes of forest fires
- Favorite sports
- Political affiliation

Hypotheses:

$H_0$  : The distribution is as claimed.

$H_1$  : The distribution is different.

# Degrees of Freedom for Goodness-of-Fit

Formula:

$$df = n - 1$$

where  $n$  is the number of categories.

Example:

Four cereal brands

$$df = 4 - 1 = 3$$

# Test of Independence

Use this test when there are two categorical variables.

Example:

|           | Yes | No  | No Opinion |
|-----------|-----|-----|------------|
| Borough B | 463 | 86  | 31         |
| Borough M | 295 | 189 | 36         |

Hypotheses:

$H_0$  : The variables are independent.

$H_1$  : The variables are dependent.

# Degrees of Freedom for Independence

Formula:

$$df = (R - 1)(C - 1)$$

where

- $R$  = number of rows
- $C$  = number of columns

Example:

$$df = (2 - 1)(3 - 1) = 2$$

# Chi-Square Tests Are Always Right-Tailed

Students should memorize:

Every Chi-Square Test is Right-Tailed.

The rejection region is always located in the right tail.



# What the Final Exam Will Ask

For every Chi-Square question:

- 1 State  $H_0$  and  $H_1$ .
- 2 Identify the claim.
- 3 Find the degrees of freedom.
- 4 State that the test is right-tailed.
- 5 Find the p-value.
- 6 Make a decision.
- 7 Write a conclusion.

## Goodness-of-Fit

- 1 Enter observed counts in  $L_1$ .
- 2 Enter expected counts in  $L_2$ .
- 3 STAT
- 4 TESTS
- 5  $\chi^2$ -GOF Test
- 6 Read the p-value.

## Test of Independence

- 1 Enter the contingency table in Matrix A.
- 2 STAT
- 3 TESTS
- 4  $\chi^2$ -Test
- 5 Read the p-value.

# Recipe-Style Goodness-of-Fit Example

Healthy Food Company cereal:

| Brand | A   | B   | C   | D   |
|-------|-----|-----|-----|-----|
| Count | 212 | 284 | 254 | 250 |

$$\alpha = 0.05$$

## Step 1

$H_0$  : The distribution is as claimed

$H_1$  : The distribution is different

## Step 2

$$df = 4 - 1 = 3$$

### Step 3

Use the TI-84  $\chi^2$ -GOF Test.

### Step 4

Read the p-value.

### Step 5

Compare the p-value with  $\alpha$ .

### Step 6

Write a conclusion in context.

## Review Worksheet Question 26

A survey produced the following results.

|           | Yes | No  | No Opinion |
|-----------|-----|-----|------------|
| Borough B | 463 | 86  | 31         |
| Borough M | 295 | 189 | 36         |

Use

$$\alpha = 0.01$$

to test whether borough and opinion are independent.

## Step 1: Hypotheses

$H_0$  : Borough and opinion are independent

$H_1$  : Borough and opinion are dependent

## Step 2: Identify the claim

The claim is  $H_0$ .

## Step 3: Degrees of freedom

$$df = (2 - 1)(3 - 1) = 2$$

## Step 4: Tail

All chi-square tests are right-tailed.

### Step 5: Calculator

Use the TI-84  $\chi^2$ -Test.

Suppose the calculator gives

$$p = 1.32 \times 10^{-16}$$

### Step 6: Decision

$$p < \alpha$$

$$1.32 \times 10^{-16} < 0.01$$

Reject  $H_0$ .

### Step 7: Conclusion

There is sufficient evidence that borough and opinion are dependent.

# Summary

For the Final Exam:

- Identify the Chi-Square test.
- State  $H_0$  and  $H_1$ .
- Find the degrees of freedom.
- Chi-Square tests are always right-tailed.
- Use the TI-84 to obtain the p-value.
- Compare the p-value with  $\alpha$ .
- Write a conclusion in context.

Good Luck!